

From Firms to Transaction-Scoped Institutions: A Coasean Re-Examination Under Costless Bilateral Enforcement

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Abstract

Coase [1937] argued that firms exist because the cost of discovering prices, negotiating terms, and enforcing contracts in open markets sometimes exceeds the cost of hierarchical management. Williamson [1985] refined this into Transaction Cost Economics (TCE), identifying asset specificity, uncertainty, and frequency as drivers of vertical integration. Hart [1995] located the firm boundary at the allocation of residual control rights over non-contractible decisions. Across these traditions, a common premise is that contract enforcement between strangers is costly, slow, and jurisdiction-dependent. This paper examines what happens to the boundary of the firm when one of those underlying costs collapses to a fixed, jurisdiction-displaced capital lockup.

We take as given the existence of a settlement primitive—developed formally in a companion paper [Daliana, 2026a], with a verified Solidity implementation in a second companion [Daliana, 2026b]—that prices the cost of trust at exactly $2\times$ the transaction value in temporarily locked capital, with no recurring extraction, no arbitrator, and no jurisdictional dependency. Treating this primitive as exogenous, we ask two questions. First, what does the existence of such a primitive imply for the Coasean threshold that defines the boundary of the firm? Second, what is the firm’s status in the organizational space once that threshold has shifted?

Our argument is that the firm has always been a coordination mechanism with a particular layer added — *subordination*: the organization of labor and assets under an ownership nucleus that retains residual control rights and captures residual output. In Coase’s analysis, the subordination layer is not incidental; it is load-bearing, because enforcement between non-subordinated counterparties was costly in the pre-primitive regime. A fixed-cost bilateral-bond settlement primitive prices that enforcement directly. The subordination layer loses its cost advantage for the class of transactions identified below; what emerges is not the abolition of the firm, but its demotion from the privileged organizational container to one pattern among several. A companion political-economy paper [Daliana, 2026c] develops what this demotion looks like from the outside; the present paper is internal institutional-economics.

Keywords: transaction cost economics, theory of the firm, subordination, coordination economics, organizational substrate, institutional economics

1 Introduction

The boundary of the firm has been the central object of theoretical attention in organizational economics since Coase [1937]. Williamson’s Transaction Cost Economics [Williamson, 1985] identified asset specificity, uncertainty, and frequency as the conditions under which hierarchical

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governance dominates market exchange. The property-rights theory [Hart, 1995] located the firm boundary at the allocation of residual control rights over non-contractible decisions. Across these traditions, a common premise is that market contracting between strangers is expensive: search is costly, negotiation is costly, and enforcement—through courts, arbitrators, or platforms—is costly, slow, and jurisdiction-dependent.

This paper asks what happens to that boundary when one of the underlying costs—contract enforcement between mutually distrusting parties—collapses to a fixed, jurisdiction-displaced capital lockup. The collapse is not a thought experiment: a companion paper [Daliana, 2026a] presents the mechanism with formal proofs of cooperation as the unique surviving Nash equilibrium, and a second companion [Daliana, 2026b] presents a verified Solidity implementation audited through TLA⁺ model checking, property-based fuzzing, symbolic execution, and specification checking.

We treat that primitive as exogenous. The contribution of this paper is institutional-economic: given the primitive, what follows for the theory of the firm? We argue two things. First, the Coasean threshold shifts inward — narrowly, in a precise sense we make explicit. Second, the firm ceases to be the privileged organizational container in the region where the threshold has shifted; it becomes one pattern among several, sharing the organizational space with treasury communities, peer networks, transaction-scoped assemblies, agent coalitions, and forms whose naming is yet to stabilize.

Paper organization. Section 2 summarizes the settlement primitive, citing the companion papers for the formal development and implementation. Section 3 develops the Coasean threshold shift. Section 4 treats the firm as one organizational pattern among many, introducing subordination as the variable whose cost advantage the primitive erases, and developing the three concrete implications for property-rights theory and for the unit of coordination. Section 5 examines token denomination as a coordination signal that supplants standing platform credentialing. Section 6 concludes.

2 The Settlement Primitive

We summarize here only the properties of the primitive that the present argument uses. The full definitions and theorems are in [Daliana, 2026a]; the implementation and verification in [Daliana, 2026b].

Asymmetric bonding. For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. Only the buyer can trigger resolution. On resolution, the buyer recovers P (net cost: $-P$) and the seller recovers $2G + P$ (net gain: $+P$). On non-resolution, both bonds remain locked indefinitely.

Equilibrium properties. In the resulting one-shot bonding game, cooperation weakly dominates defection for all parties, and the cooperative profile is the unique outcome surviving iterated elimination of weakly dominated strategies. The result extends to N -party process trees through *progressive collateralization*, with the seller’s risk-to-reward ratio growing with chain depth.

Atomic resolution and peer enforcement. All orders within a process resolve atomically—all or nothing—which induces a weakest-link subgame among sellers. This produces an endogenous coordination pressure analogous to peer enforcement in joint-liability microfinance, but obtained without repeated interaction, local information, or social punishment technology.

No escape hatches. The primitive has no timeout, no admin override, no governance mechanism, no protocol fee, and no upgrade path. The companion mechanism paper proves that any such escape hatch weakens the equilibrium.

What the primitive does *not* provide. For the institutional argument that follows, two non-properties matter as much as the properties. The primitive does not provide capital aggregation across projects, does not provide pooled risk-bearing across uncorrelated ventures, and does not provide legal personhood for off-chain liability. We return to these absences in Section 4.

3 The Coasean Threshold Shift

Coase’s central insight is that firms exist because market transaction costs—searching for partners, negotiating terms, enforcing agreements—sometimes exceed the cost of hierarchical management. Williamson refined this into TCE, identifying three conditions under which hierarchical governance dominates market exchange: high asset specificity, high uncertainty, and high frequency.

The settlement primitive changes the calculus by collapsing one of these costs to a fixed, predictable capital lockup. Consider the three transaction costs separately.

Search costs. Public on-chain signals (settlement history, accepted-token lists, attested capabilities) allow autonomous discovery through shared coordination graphs. For agents able to read blockchain state, search cost approaches zero. The substantive change is not that search is cheaper—platforms already make search cheap—but that the search index is non-proprietary: no platform owns the discovery function, and the data is not extracted as rent.

Negotiation costs. Off-chain commitment building (EIP-712 typed structured data) lets parties iterate on terms without gas costs. The negotiation is bilateral and direct; counterparties can discover and iterate through shared graph visibility rather than through a standing intermediary’s matching function. The substantive change here is asymmetric: the *cost* of negotiating is roughly unchanged from web2 baselines, but the *counterparty space* from which one can negotiate is no longer mediated by a platform’s terms of service.

Enforcement costs. This is the decisive variable. In the pre-primitive world, enforcement requires courts (slow, expensive, jurisdiction-dependent), arbitrators (trusted third parties), or standing intermediaries that internalize coordination and charge for it. The primitive prices enforcement at exactly $2\times$ the transaction value—a fixed, predictable, jurisdiction-displaced cost requiring no third party.¹

Proposition 3.1 (Coasean Threshold Shift). *For any transaction where $2P$ (the capital lockup cost) is less than the firm’s coordination overhead for the same transaction, the rational organizational choice is market exchange via the primitive rather than hierarchical integration.*

¹The $2P$ figure is the capital-lockup component, which is the substantive change relative to the prior regime. A full enforcement-cost accounting adds (i) gas for the `commit` and `resolveProcess` calls (bounded but non-zero), (ii) opportunity cost on locked capital over the process duration T at the prevailing risk-free rate r (roughly $2P \cdot rT$), (iii) slippage if bond-denomination currency must be acquired at transaction time, and (iv) operational-security costs associated with key custody (signing infrastructure, backup, rotation). The Coasean-threshold comparison below abstracts (i)–(iv) because they are small relative to the standing-firm overheads against which the comparison is made for the retail-scale transactions this paper treats; the abstraction is not universally valid and should be revisited for applications in which any of (i)–(iv) is comparable in magnitude to $2P$.

Theorem 3.1 is a comparative-statics statement, not a sweeping institutional prediction. It identifies a class of transactions for which the make-or-buy boundary tilts toward *buy*—specifically, transactions where the time-weighted opportunity cost of locked capital is lower than the standing overhead of internalizing the coordination function. The class is non-empty (most retail-scale transactions in mature financial environments satisfy it) and bounded (very-high-frequency, very-low-margin, or very-long-duration transactions do not). The proposition does not claim that all transactions migrate; it claims that the migration threshold is now defined by capital opportunity cost rather than by the cost of standing institutional infrastructure.

Engagement with TCE. Williamson’s three drivers of vertical integration deserve direct treatment. *Asset specificity* continues to favor hierarchy: a specialized asset whose value is contingent on a specific counterparty cannot be efficiently coordinated through a sequence of short-lived bonded transactions, since the holdup risk is in the asset’s pre-transaction sunk investment, not in the transaction itself. *Uncertainty* cuts both ways: contract incompleteness favors hierarchy, but the primitive’s evidence trail (timestamped commitments, signed disclosures) reduces some classes of uncertainty—specifically those that hierarchical authority historically resolved by managerial fiat. *Frequency* is the condition under which the shift is most pronounced: high-frequency exchanges with low asset specificity are exactly where the standing firm’s overhead is hardest to amortize and the primitive’s per-transaction capital lockup is easiest to absorb.

The proposition therefore narrows rather than overturns TCE. It identifies a region of the asset-specificity / uncertainty / frequency space where hierarchical governance was previously the equilibrium outcome but is no longer.

4 The Firm as One Pattern Among Many

The implication of Theorem 3.1 is not that the firm dissolves but that the firm’s status in the organizational space changes. The firm has always been a coordination mechanism. What distinguishes the modern corporate firm from older coordination forms—partnerships, guilds, cooperatives, market exchange—is not the coordination function itself but a particular overlay placed on top of it: *subordination*.

4.1 Subordination as the Variable

By subordination we mean the allocation of labor and productive assets under a single ownership nucleus, combined with an employment contract that transfers residual output rights and residual control rights to that nucleus in exchange for a wage. In the Coasean analysis, subordination is not incidental to the firm; it is load-bearing. Prior to the settlement primitive, enforcement between non-subordinated counterparties was costly, and aggregation under an ownership nucleus that resolved enforcement internally through managerial authority was the institutional response.

Proposition 4.1 (Subordination as an Enforcement Economy). *In the pre-primitive regime, subordination is a response to enforcement cost. The firm’s subordination layer secures coordination between counterparties whose direct bilateral enforcement would be prohibitively costly, at the price of aggregating residual rights under a single nucleus. When bilateral enforcement is priced at a fixed capital lockup, the subordination layer loses its cost advantage in the region identified by Theorem 3.1.*

Theorem 4.1 reframes what is dissolving. It is not the firm’s coordination function that becomes unnecessary — coordination is still required. It is the subordination overlay — the transfer of residual control rights from the producing party to an ownership nucleus — that

becomes economically optional where the primitive is applicable. What is left behind is coordination without subordination: the same productive assets, the same coordination requirements, but without the ownership aggregation that historically secured them.

4.2 What Emerges in the Shifted Region

Where subordination is no longer load-bearing, the firm becomes one organizational pattern among several. The organizational space in that region admits multiple viable coordination patterns that were previously suppressed by the enforcement-cost structure: transaction-scoped assemblies that form around a process and dissolve at settlement; standing treasury communities whose members recurrently participate in compatible process trees; peer networks of independent bonded counterparties without a common treasury; agent coalitions mixing human, software, and machine participants. The firm continues to exist; it shares the space with these other forms rather than occupying it alone. Which form dominates for a given industry is an empirical question about the parameters of Theorem 3.1 and the non-coordination functions (capital aggregation, risk pooling, legal personhood) that the primitive does not itself address.

Holacracy as an intra-firm precedent. Intra-firm experiments with flat coordination — most visibly the Zappos adoption of Holacracy [Robertson, 2015] in 2014 — previewed what some of these patterns look like when they remain confined to a single ownership nucleus. Such experiments are informative as data points about what flat coordination requires operationally, but they are not what the primitive generalizes. Holacracy, once the subordination nucleus is retained, operates at the mercy of that nucleus. The primitive enables flat coordination where the nucleus itself is not required; that is a stronger condition than any intra-firm reform can provide.

4.3 Three Concrete Implications

(a) The wallet is the unit of coordination. Each productive asset is represented directly by a wallet address. At the protocol tier² the wallet can hold content-addressed credentials — attested capabilities, operator profiles, and (at the runtime tier) NFT-anchored deeds — that certify the asset’s identity and performance history. In the firm model, the cook’s skill, the kitchen’s equipment, the courier’s vehicle are subsumed under the firm’s legal identity and cease to be individually addressable. In the wallet model, each asset is its own coordination endpoint. The productive asset *is* the identity; it remains an identity whether the firm persists or not.

(b) Tokens replace stock. Participants in the shifted region do not receive shares in a firm; they hold tokens denominated by the communities with which they coordinate. Token holdings signal community affiliation and participation-history, not residual-output ownership in a centralized nucleus. The relationship between productive asset and community is market-mediated rather than hierarchically internalized. Tokens are borderless, community-defined rather than jurisdiction-issued, and the community’s definition is itself an open parameter.

(c) Actor type is neutral. Nothing in the primitive’s coordination layer distinguishes an AI capable of bearing capital risk from a human capable of the same risk. A person with a law degree and an AI trained on legal precedent compete on identical terms — each as a bonded counterparty whose capabilities, if attested at all, are attested at the protocol tier by a schema-validated attestation coordinator rather than asserted at the kernel tier. Actor-type neutrality

²We use “kernel / protocol / runtime” in the sense defined in [Daliana, 2026a, §1.4]. The kernel guarantees only the `commit/resolveProcess` bond structure; the credential and attestation apparatus described in this paragraph is realized at the protocol tier by extensions (attestation coordinators, schema registries) whose composition discipline is developed in the implementation companion [Daliana, 2026b].

is a property of the kernel’s coordination layer, not a policy choice; it follows directly from the kernel’s indifference to what sits behind a wallet.

4.4 Engagement with the Property-Rights Tradition

The property-rights tradition [Hart, 1995, Grossman and Hart, 1986, Hart and Moore, 1990] locates the firm boundary at the allocation of residual control rights over non-contractible decisions. The primitive does not abolish non-contractibility; rather, it changes *where* residual control rights are allocated. In the firm model, those rights are allocated to the ownership nucleus. In the wallet model, they are allocated to whoever controls the wallet — which, under self-sovereign key management and NFT-anchored credentials, is the productive asset itself (or the individual operating it). Residual control rights do not disappear; they are held by the asset rather than aggregated under an ownership layer.

This is the institutional-economics translation of what Section 4 has been describing throughout: the decision about who holds residual rights, previously answered by the firm’s ownership structure, is now a property of the asset’s wallet architecture. This relocates the allocation question rather than dissolving it; Hart and Moore [1990] applies to the wallet-layer question just as it applied to the firm-layer question before, with a different set of empirical facts about costs and observability.

Williamson’s drivers, revisited. Our treatment of Williamson’s three drivers (Section 3) likewise does not dissolve TCE; it relocates the question. Asset specificity continues to matter wherever a productive asset’s value is contingent on a specific bond-specific investment; the primitive does not change that. What changes is that the institutional response to asset specificity need not be subordination under an ownership nucleus — a treasury community, a long-term peer network, or a serial bonded-commitment pattern can, for some parameter regions, secure the specific investment without aggregating residual rights.

The non-coordination functions of the firm — capital aggregation, organizational knowledge, risk pooling, legal personhood — persist as durable requirements that some institution must satisfy. They do not necessarily re-bundle into a firm-shaped container; they can be satisfied by treasuries, DAOs, mutuals, cooperatives, or jurisdiction-specific legal wrappers attached to wallet clusters. The empirical question of which bundling pattern dominates in a given industry is outside the scope of this paper.

The firm’s self-knowledge function dissolves alongside. A function we have not separately named is that the firm historically has been the unit at which *books* are kept — the unit whose position is tracked over time through double-entry bookkeeping in the tradition of Pacioli [1494]. The bookkeeping function is not incidental: at scale, an entity that cannot maintain accurate records of its position cannot grow, attract investment, pay taxes, or contract with third parties who require disclosure. The companion accounting paper [Daliana, 2026g] develops the argument that a FIGARO process is a self-closing ledger period, and that the apparatus that grew up around the books (bookkeeping, accounting, audit) becomes optional for activity that settles through the protocol. The implication for the present argument is that the firm’s demotion from privileged organizational container is supported by an additional structural shift not addressed elsewhere above: the records function that motivated the firm boundary historically is also no longer firm-bound. A wallet cohort participating in many process trees has its full position visible without consolidation.

5 Token Denomination as Coordination Signal

In the primitive, the choice of denomination token is itself a coordination signal. The settlement layer is token-agnostic: any ERC-20 can serve as the bond and payment currency. This means:

- A stablecoin signals legal-system alignment.
- A DAO governance token signals community membership.
- A commodity-backed token signals value anchoring.
- Accepting a specific token *is* the seller’s identity declaration—an accepted-token list is a brand.

Settlement velocity in a given token replaces traditional reputation metrics (star ratings, review scores). The on-chain record of resolved orders denominated in token τ is a public, tamper-proof signal of the seller’s reliability within the τ economy. No platform needs to aggregate, curate, or attest this signal; it is a natural byproduct of protocol usage.

This matters institutionally because reputation infrastructure is one of the durable rents that platforms extract. A standing platform’s credentialing function—certifying that a counterparty is reliable, safe, or qualified—is one of the structural reasons platforms persist even after their matching function is technically replicable. When reputation is a public byproduct of settlement rather than a proprietary platform asset, the platform’s claim to that rent weakens.

Schelling-point tokens as a distinct category. The coordination-signal argument above applies to *any* token used in a bonded commitment. A second coordination role is played by tokens designed specifically to serve as focal points in the sense of Schelling [1960]: anchors around which participants converge to signal alignment with a specific ecosystem, without central authority and without relying on any of the token’s financial-mechanism features. The cryptoeconomics companion [Daliana, 2026f] develops this category in detail, with BTC and ETH as precedents at the money and settlement layers respectively, and the FIG token as the focal anchor for participants who coordinate around the bonded-commitment primitive specifically. A Schelling-point token does not compete with the general-purpose denomination tokens discussed above; it carries a signal they cannot carry, which is alignment with a specific coordination ecosystem rather than with legal-system, community, or value-anchoring choices. Such a token is underpinned not by cost-of-production (as Bitcoin is) nor by transaction-execution demand (as Ethereum is) but by the *value-added of trade* conducted through the substrate it coordinates around: FIG’s value rests on the scale and quality of bonded commitments conducted by participants who use it to identify each other.

6 Conclusion

The Coasean threshold is not a conjecture—it is observable. Contemporary coordination stacks often extract 15–30% of transaction value as coordination rent (Uber 25%, DoorDash 30%, Airbnb 14%, Amazon Marketplace 15%). Credit card networks layer 2–3% interchange plus double-digit annual interest on revolving balances. Financial markets intermediate trillions in notional value against a fraction of that in real economic exchange. Asymmetric bonding prices the cost of trust at 2× the transaction value in temporarily locked (not spent) capital, with zero recurring extraction.

For any economic activity where the coordination rent exceeds the opportunity cost of locked capital, the standing firm — as Coase conceived it — is no longer the uniquely efficient unit of economic organization in that activity. The stronger consequence is that the firm becomes one organizational pattern among several in the shifted region: transaction-scoped assemblies,

treasury communities, peer networks, and agent coalitions share the coordination space alongside it. The settlement primitive does not abolish coordination, nor does it abolish the firm; it retires the *subordination overlay* that previously made the firm the privileged container, and leaves a richer organizational space in its place.

The argument leaves three questions open. First, the empirical question: which industries and which transaction classes lie above the threshold today, and at what rate does the threshold shift as on-chain settlement matures? Second, the institutional question: which organizational patterns — firms, treasuries, peer networks, or forms not yet named — dominate in a given industry once the primitive is available, and what determines the dominance? The non-coordination functions of the firm (capital aggregation, organizational knowledge, risk pooling, legal personhood) must still be bundled somewhere; the bundling question is open. Third, a question of a different character: why this shift is qualitatively harder to recognize than to specify, and what that difficulty itself reveals about the architecture of contemporary coordination infrastructure. That third question is addressed in the companion political-economy paper [Daliana, 2026c].

A note on scope alignment across the companion papers. This paper bounds its claim by two scope conditions: the Coasean comparison (Theorem 3.1: $2P$ vs. coordination overhead) and asset specificity (Section 4). The political-economy companion [Daliana, 2026c] bounds its claim differently, by hegemonic legibility — its argument applies wherever the primitive makes a previously-presupposed coordination function visible as a design artifact. The labor-law companion [Daliana, 2026e] bounds its claim by asset specificity alone. The electronic-evidence companion [Daliana, 2026d] bounds its claim least restrictively — wherever the primitive is used, its evidentiary framework applies. Each bound is internally reasonable within its paper’s discipline. Read across the set, these bounds are compatible but not identical. The theoretical-cluster domain is the *intersection* of the four scope conditions, not their union: a given transaction lies within the cluster’s shared domain only if the primitive’s Coasean threshold is met, asset specificity is low or structurally handled, the coordination function is legible as a design choice, and the evidentiary framework is practically accessible for the parties’ jurisdictional situation. A transaction that satisfies three of the four conditions but not the fourth is in-scope for three papers and out-of-scope for one; readers constructing arguments about such transactions should check which papers apply. We flag this here because each companion’s scope condition is developed within its own register and the inter-paper alignment is not elsewhere made explicit.

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